

Important decisions

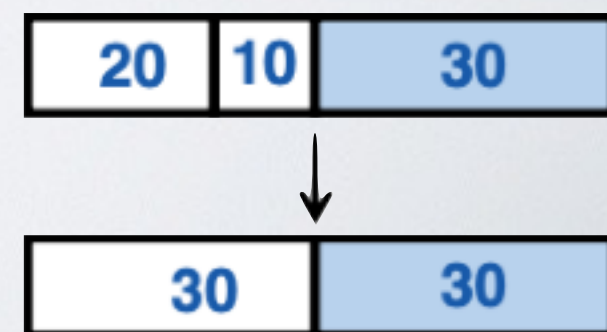
Placement choice : where in free memory to put a requested block?

- Freedom : can select any memory in the heap
- Ideal : put block where it won't cause fragmentation later (impossible in general, requires future knowledge)

Split free blocks to satisfy smaller requests?

- Freedom : can choose any larger block to split
- Ideal : choose block to minimize fragmentation

Coalescing free blocks to yield larger blocks



Fragmentation is impossible to solve

Theoretical result

For any allocation algorithm, there exist streams of allocation and deallocation requests that defeat the allocator and force it into severe fragmentation

➡ Avoiding fragmentation is impossible